

ANSWER KEY

Module:	Principles of Machine Learning		
Module Code	QHM5703	Paper	A
Time allowed	2 H	Filename	QHM5703 PML First Sit
Rubric	ANSWER ALL FOUR QUESTIONS		
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Solution Key

Question 1

- a) In this question we explore regression in a problem involving two attributes x and y , where x is the predictor feature and y is the prediction. The dataset that we will use for training contains four samples and is shown in Table 1.

x	y
2	1
4	5
1	2
3	2

Table 1

The coefficients $\mathbf{w}$ of the Minimum Mean Square Error (MMSE) solution of a simple linear model can be obtained as

$$\mathbf{w} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

where $\mathbf{X}$ is the design matrix and $\mathbf{y}$ is the prediction vector.

[15 marks]

- (i) Obtain the MMSE coefficients of the simple linear model $y = w_0 + w_1 x$. You can use the following intermediate result:

$$(\mathbf{X}^T \mathbf{X})^{-1} = \begin{bmatrix} 1.5 & -0.5 \\ -0.5 & 0.2 \end{bmatrix}$$

(10 marks)

Answer Key [solution should contain following key-points]:

Design Matrix: $\mathbf{X}$

$$\mathbf{X} = \begin{bmatrix} 1 & 2 \\ 1 & 4 \\ 1 & 1 \\ 1 & 3 \end{bmatrix}$$

$$\mathbf{X}^T = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 4 & 1 & 3 \end{bmatrix}$$

$$\mathbf{X}^T \mathbf{X} = \begin{bmatrix} 4 & 10 \\ 10 & 30 \end{bmatrix}$$

$$(X^T X)^{-1} = \begin{bmatrix} 1.5 & -0.5 \\ -0.5 & 0.2 \end{bmatrix}$$

$$y = \begin{bmatrix} 1 \\ 5 \\ 2 \\ 2 \end{bmatrix}$$

$$X^T y = \begin{bmatrix} 10 \\ 30 \end{bmatrix}$$

$$w = (X^T X)^{-1} X^T y = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

- (ii) Calculate the training Mean Square Error (MSE) of the MMSE solution (Use $w_0 = 0$ and $w_1 = 1$ if you did not obtain the MMSE solution).

(5 marks)

Answer Key [solution should contain following key-points]:

$$MSE = \frac{1}{N} \sum_1^N (y_i - f(x_i))^2$$

Errors e : 1, -1, -1, 1

Squares. e^2 : 1, 1, 1, 1

$$MSE = 4/4 = 1$$

- b) Consider the cubic model $y = w_0 + w_1x + w_2x^2 + w_3x^3$ for the dataset in Table1.

[10 marks]

- (i) What would you expect the training MSE of this model to be?

(4 marks)

Answer Key [solution should contain following key-points]:

The training MSE would be zero, as we can always find a polynomial of order 4 that goes through the 4 points.

- (ii) Assuming that the true model is $y = x + n$, where n is zero-mean Gaussian noise, identify the main sources of error in the prediction of this cubic model during deployment.

(6 marks)

Answer Key [solution should contain following key-points]:

The main sources of error would be the variance of the random component n , the model bias, i.e. the difference between the true pattern (line) and the predicted pattern (cubic model), and the model variance due to sampling.

Solution Key

Question 2

- a) In this question we explore classification in a problem involving two predictor features x_A and x_B , and two classes, namely $\circ$ (positive class) and $\times$ (negative class). The dataset that will be used in this question is shown in Figure 1.

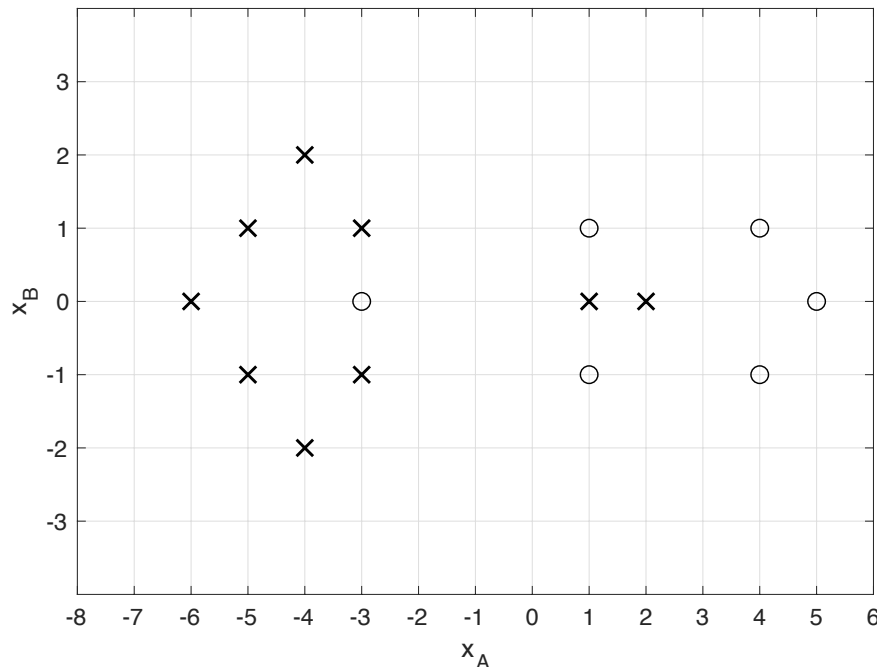


Figure 1

Consider a family of linear classifiers defined by $\mathbf{w}^T \mathbf{x} = 0$, where $\mathbf{w} = [w_0, w_A, w_B]^T$ and $\mathbf{x} = [1, x_A, x_B]^T$. A sample $\mathbf{x}_i$ such that $\mathbf{w}^T \mathbf{x}_i > 0$ will be labelled as $\circ$, otherwise it will be labelled as $\times$. Given the linear classifier defined by the coefficients $w_B = 0$, $w_A = 1$ and $w_0 = 2$:

[10 marks]

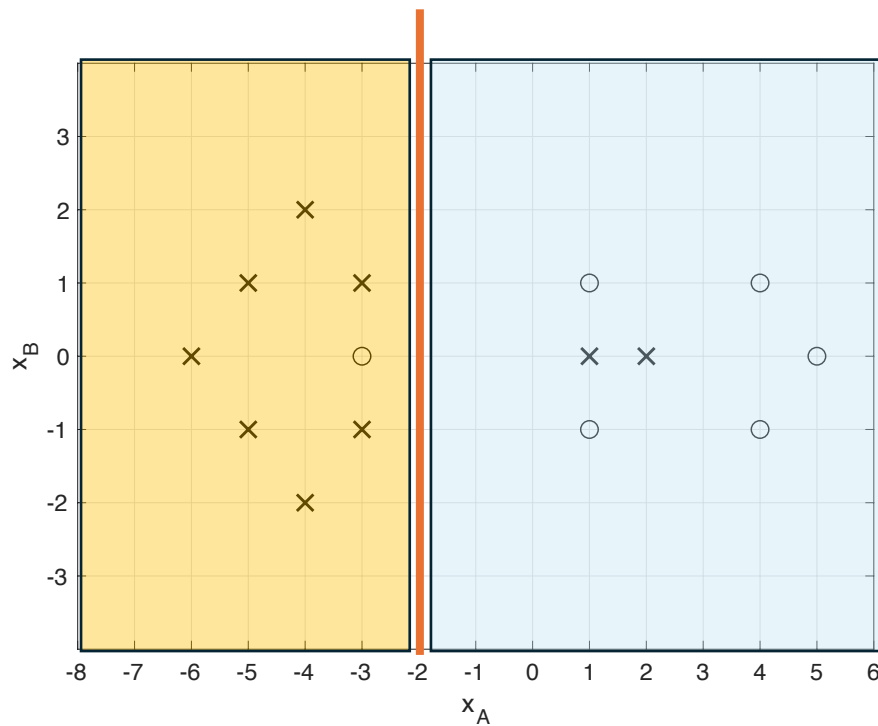
- (i) Obtain the classifier's decision regions.

(4 marks)

Answer Key [solution should contain following key-points]:

Drawing of the boundary

The resulting boundary is $x_A = -2$, hence the half-plane $x_A > -2$ corresponds to the $\circ$ class, and $x_A < -2$ to the $\times$ class.



- (ii) Obtain the classifier's confusion matrix for the dataset shown in Figure 1 and identify its sensitivity and specificity.

(6 marks)

Answer Key [solution should contain following key-points]:

Showing a confusion matrix. Each correct value, total of 4 values.

		Actual	
		○	×
Predicted	○	5	2
	×	1	7

Computing sensitivity = ?

Specificity = ?

- b) Assume that we want to build a Bayes classifier that uses as predictor feature x_A .

[15 marks]

- (i) Obtain the priors for each class, namely $P(\circ)$ and $P(\times)$, and the means of the distributions $P(x_A | \circ)$ and $P(x_A | \times)$.

(8 marks)

Answer Key [solution should contain following key-points]:

The probability of each class is $P(\bigcirc)=6/15$ and $P(\times)=9/15$. The mean of class $\bigcirc$ is **-3** and the mean of class $\times$ is **2**.

(ii) Describe the corresponding Bayes classifier.

(4 marks)

Answer Key [solution should contain following key-points]:

The classifier will compute $P(x_A|\bigcirc) P(\bigcirc)$ and $P(x_A|\times) P(\times)$ and will assign the label corresponding to the class with the highest probability.

(iii) If the standard deviations of $P(x_A | \bigcirc)$ and $P(x_A | \times)$ are equal, how would a sample such that $x_A = -0.5$ be classified?

(3 marks)

Answer Key [solution should contain following key-points]:

This sample is halfway between the means of both classes. Since the variances are equal $P(x_A | \bigcirc) = P(x_A | \times)$, it will assign the label of the class with the highest probability, i.e to $\times$.

Question 3

- a) This question concerns model optimisation in machine learning. The k-means algorithm is said to converge to local minima, rather than to the global minimum.

[15 marks]

- (i) Use the notion of error function to explain the concepts of local minimum and global minimum.

(5 marks)

Answer Key [solution should contain following key-points]:

The error function associates an error value to each candidate model. A local minimum refers to any model which provides the lowest error within a vicinity, whereas a global minimum is the model with the lowest error amongst all the candidate models.

- (ii) Explain what is meant by the statement *the k-means algorithm converges to a local minimum*.

(5 marks)

Answer Key [solution should contain following key-points]:

The k-means algorithm defines an iterative process where the solution is updated to models with lower cost. Convergence to a local minimum implies that the model provided by k-means will be the best one nearby the initial model, not necessarily the global minimum.

- (iii) Considering the risk of converging to a local minimum, design a strategy that would improve the solution provided by the k-means algorithm.

(5 marks)

Answer Key [solution should contain following key-points]:

Since the k-means algorithm always converges to the best solution close to the initial point, a method to improve its performance would consist of restarting from different random initial points the algorithm and selecting the best solution.

- b) The validation-set approach allows one to evaluate the performance of different models during model selection.

[10 marks]

- (i) After applying a validation-set approach, the validation errors of two models f_1 and f_2 are found to be respectively $E_1 = 10$ and $E_2 = 12$. How would you use this result to inform your selection of either f_1 or f_2 ?

(4 marks)

Answer Key [solution should contain following key-points]:

In principle the second model shows a better deployment performance, however this difference could be due by chance, so it would be necessary to evaluate the significance of this difference.

- (ii) Due to the low number of samples in the available dataset, it is suggested that the whole dataset should be used for training models f_1 and f_2 and both models should be compared based on their training errors. What is your view on this suggestion?

(6 marks)

Answer Key [solution should contain following key-points]:

Training errors should never be used for assessing the performance of a model. Training errors do not evaluate the ability of a model to generalise, hence it is not a valid metric.

Solution Key

Question 4

a) This question concerns neural networks.

[15 marks]

(i) Which layers offer greater flexibility, fully-connected layers or convolutional layers?

(5 marks)

Answer Key [solution should contain following key-points]:

In fully connected layers all the input features are associated to all perceptrons and have different weights. Convolutional layers can be seen as a specific case of fully connected layers, where the weights defined by a filter are shared by all the perceptrons. Hence fully-connected layers are more flexible. From the point of view of number of parameters, it can also be concluded that fully-connected layer has more parameters compared to convolutional layer, thus it has more flexibility.

(ii) Why are convolutional networks suitable for time series and image data?

(5 marks)

Answer Key [solution should contain following key-points]:

Time series and image data define topological relations between input features and can exhibit the property of equivariance to translation, which is captured by the convolution operation.

(iii) The number of feature maps in convolutional architectures usually increase as we move closer to the output layer. What is the idea behind this design?

(5 marks)

Answer Key [solution should contain following key-points]:

Convolutional architectures define a processing pipeline where raw features are processed producing new features or concepts along the way. By having many feature maps at the end, we expect to produce complex concepts useful for the task at hand. As for the spatial or temporal location of the features becomes less important and presence of complex features, for the tasks such as classification becomes more important towards the end.

- b) Consider a dataset consisting of grayscale images of size 100×100 pixels and a binary label. A deep neural network combining convolutional, pooling and fully-connected layers is chosen for building a classifier for this dataset.

[10 marks]

- (i) The first hidden layer is a convolutional one and consists of two 100×100 feature maps. Each map is obtained by applying a different filter of dimensions 3×3 . How many parameters need to be trained in the first layer?

(4 marks)

Answer Key [solution should contain following key-points]:

The number of parameters in a convolutional layer correspond to the parameters of each filter. Since there are 2 filters and each one consists of 9 parameters, the total is 18 for filters and 2 parameters for bias (one each): total = 20.

- (ii) The second layer is a 2×2 max-pooling layer. How many feature maps does this layer have and what are their dimensions?

(4 marks)

Answer Key [solution should contain following key-points]:

The number of feature maps is the same as in the previous layer, i.e. 2. 2×2 pooling provides 1 value per 2×2 region hence the dimension for each feature map is 50×50 .

- (iii) The third hidden layer is also convolutional and consists of 8 feature maps defined by filters of dimensions $3 \times 3 \times D$. What is the value of D ?

(2 marks)

Answer Key [solution should contain following key-points]:

D is the number of feature maps in the previous layer, i.e 2.